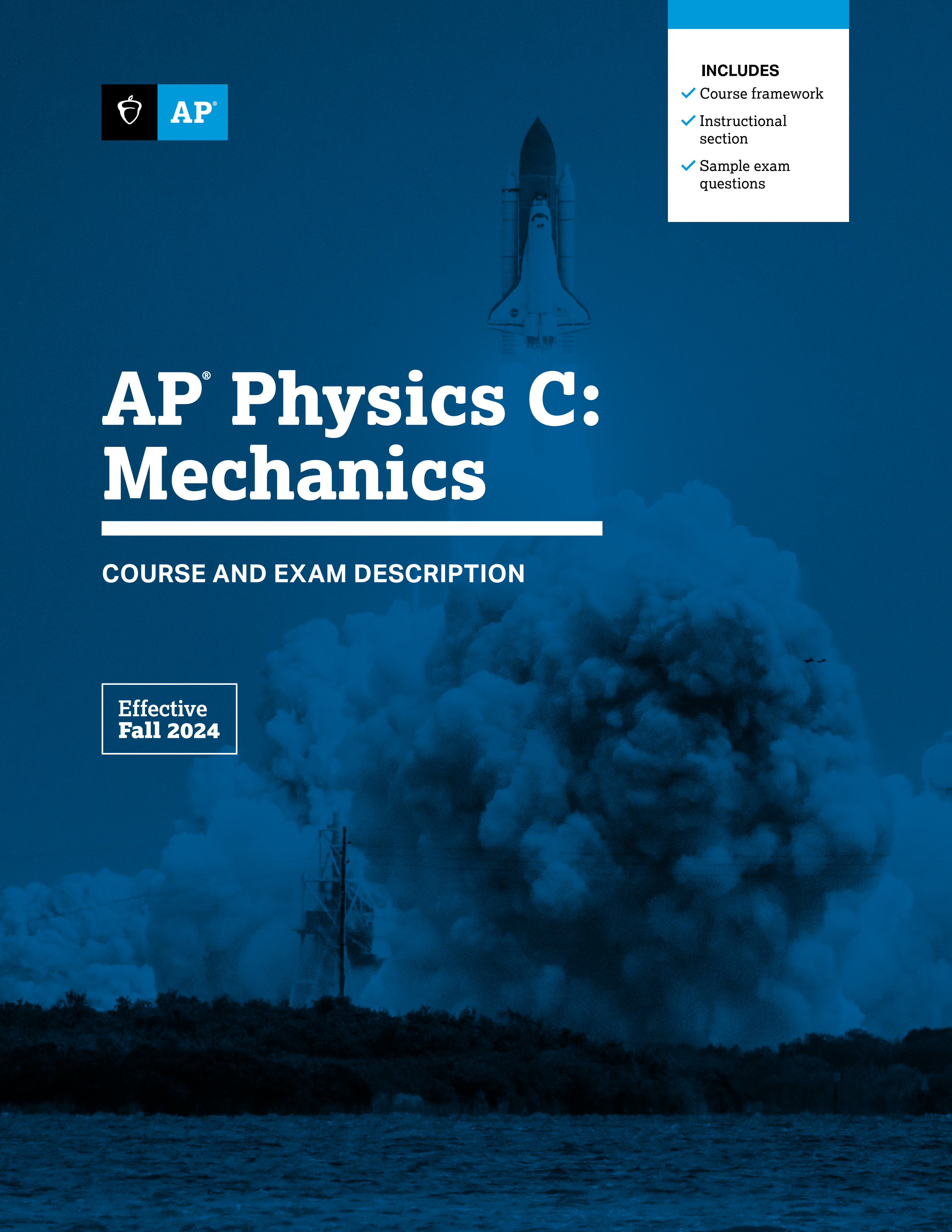


INCLUDES
- Course framework
- Instructional section
- Sample exam questions

AP® Physics C: Mechanics

COURSE AND EXAM DESCRIPTION

Effective
Fall 2024

ADVANCED PLACEMENT PHYSICS C: MECHANICS TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS
Universal gravitational constant, $G = 6.67 \times 10^{-11}\ \text{m}^3/(\text{kg} \cdot \text{s}^2) = 6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2$
Acceleration due to gravity at Earth's surface, $g = 9.8\ \text{m/s}^2$
Magnitude of the gravitational field strength at the Earth's surface, $g = 9.8\ \text{N/kg}$

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

UNIT SYMBOLS				
	hertz,	Hz	newton,	N
	joule,	J	second,	s
	kilogram,	kg	watt,	W
	meter,	m		

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin\theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan\theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following assumptions are used in this exam.

- The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- Air resistance is assumed to be negligible unless otherwise stated.
- Springs and strings are assumed to be ideal unless otherwise stated.

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0}t + \frac{1}{2}a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x\left(x - x_0\right)$$

$$\Delta x = \int v_x\left(t\right)dt$$

$$\Delta v_x = \int a_x\left(t\right)dt$$

$$\vec{x}_{\text{cm}} = \frac{\sum m_i \vec{x}_i}{\sum m_i}$$

$$\vec{r}_{\text{cm}} = \frac{\int \vec{r}\,dm}{\int dm}$$

$$\lambda = \frac{d}{d\ell}m\left(\ell\right)$$

$$\vec{a}_{\text{sys}} = \frac{\sum \vec{F}}{m_{\text{sys}}} = \frac{\vec{F}_{\text{net}}}{m_{\text{sys}}}$$

$$\left|\vec{F}_g\right| = G\frac{m_1 m_2}{r^2}$$

$$\left|\vec{F}_f\right| \leq \left|\mu \vec{F}_N\right|$$

$$\vec{F}_s = -k\Delta\vec{x}$$

$$a_c = \frac{v^2}{r} = r\omega^2$$

$$T = \frac{1}{f}$$

$$K = \frac{1}{2}mv^2$$

$$W = \int_a^b \vec{F}\cdot d\vec{r}$$

$$\Delta K = \sum W_i = \sum F_{\|,i}d_i$$

$$\Delta U = -\int_a^b \vec{F}_{\text{cf}}(r)\cdot d\vec{r}$$

$$F_x = -\frac{dU(x)}{dx}$$

$$U_s = \frac{1}{2}k\left(\Delta x\right)^2$$

$$U_G = -G\frac{m_1 m_2}{r}$$

$$\Delta U_g = mg\Delta y$$

a = acceleration
E = energy
f = frequency
F = force
h = height
J = impulse
k = spring constant
K = kinetic energy
ℓ = length
m = mass
M = mass
p = momentum
P = power
r = radius, distance, or position
t = time
T = period
U = potential energy
v = velocity or speed
W = work
x = position or distance
y = height
λ = linear mass density
μ = coefficient of friction

$$P_{\text{avg}} = \frac{W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

$$P_{\text{inst}} = \frac{dW}{dt}$$

$$\vec{p} = m\vec{v}$$

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \int_{t_1}^{t_2} \vec{F}_{\text{net}}\left(t\right)dt = \Delta\vec{p}$$

$$\vec{v}_{\text{cm}} = \frac{\sum \vec{p}_i}{\sum m_i} = \frac{\sum m_i \vec{v}_i}{\sum m_i}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt}$$

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\left(\theta - \theta_0\right)$$

$$v = r\omega$$

$$a_T = r\alpha$$

$$\vec{\tau} = \vec{r}\times\vec{F}$$

$$I_{\text{tot}} = \sum I_i = \sum m_i r_i^2$$

$$I = \int r^2\,dm$$

$$I' = I_{\text{cm}} + Md^2$$

$$\alpha_{\text{sys}} = \frac{\Sigma\tau}{I_{\text{sys}}} = \frac{\tau_{\text{net}}}{I_{\text{sys}}}$$

$$K_{\text{rot}} = \frac{1}{2}I\omega^2$$

$$W = \int \tau\cdot d\theta$$

$$\vec{L} = \vec{r}\times\vec{p} = I\vec{\omega}$$

$$\Delta L = \int \tau\,dt$$

$$\Delta x_{\text{cm}} = r\Delta\theta$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi\sqrt{\frac{m}{k}}$$

$$T_p = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T_{\text{phys}} = 2\pi\sqrt{\frac{I}{mgd}}$$

$$x = x_{\text{max}}\cos\left(\omega t + \phi\right)$$

a = acceleration
d = distance
f = frequency
F = force
I = rotational inertia
k = spring constant
K = kinetic energy
ℓ = length
L = angular momentum
m = mass
M = mass
p = momentum
r = radius, distance, or position
t = time
T = period
v = velocity or speed
W = work
x = position or distance
α = angular acceleration
θ = angle
τ = torque
ϕ = phase angle
ω = angular frequency or angular speed

GEOMETRY AND TRIGONOMETRY

Rectangle
$A = bh$

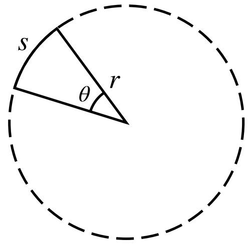
Triangle
$A = \frac{1}{2}bh$

Circle
$A = \pi r^2$
$C = 2\pi r$
$s = r\theta$

Rectangular Solid
$V = \ell wh$

Cylinder
$V = \pi r^2 \ell$
$S = 2\pi r\ell + 2\pi r^2$

Sphere
$V = \frac{4}{3}\pi r^3$
$S = 4\pi r^2$



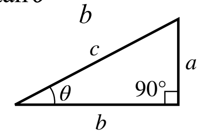
A = area
b = base
C = circumference
h = height
ℓ = length
r = radius
s = arc length
S = surface area
V = volume
w = width
θ = angle

Right Triangle
$a^2 + b^2 = c^2$

$\sin\theta = \frac{a}{c}$

$\cos\theta = \frac{b}{c}$

$\tan\theta = \frac{a}{b}$

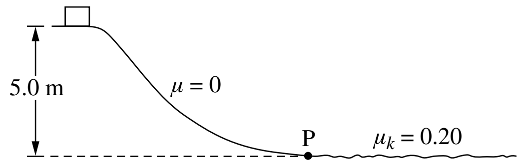


VECTORS	CALCULUS	IDENTITIES
$\vec{A}\cdot\vec{B} = AB\cos\theta$	$\frac{df}{dx} = \frac{df}{du}\frac{du}{dx}$	$\log\left(a\cdot b^x\right) = \log a + x\log b$
$\left\lvert\vec{A}\times\vec{B}\right\rvert = AB\sin\theta$	$\frac{d}{dx}\left(x^n\right) = nx^{n-1}$	$\sin^2\theta + \cos^2\theta = 1$
$\vec{r} = \left(A\hat{i} + B\hat{j} + C\hat{k}\right)$	$\frac{d}{dx}\left(e^{ax}\right) = ae^{ax}$	$\sin\left(2\theta\right) = 2\sin\theta\cos\theta$
$\vec{C} = \vec{A} + \vec{B}$	$\frac{d}{dx}\left(\ln ax\right) = \frac{1}{x}$	$\frac{\sin\theta}{\cos\theta} = \tan\theta$
$\vec{C} = \left(A_x + B_x\right)\hat{i} + \left(A_y + B_y\right)\hat{j}$	$\frac{d}{dx}\left[\sin(ax)\right] = a\cos(ax)$	
	$\frac{d}{dx}\left[\cos(ax)\right] = -a\sin(ax)$	
	$\int x^n\,dx = \frac{1}{n+1}x^{n+1}, n \neq -1$	
	$\int e^{ax}\,dx = \frac{1}{a}e^{ax}$	
	$\int \frac{dx}{x+a} = \ln\lvert x+a\rvert$	
	$\int \cos(ax)\,dx = \frac{1}{a}\sin(ax)$	
	$\int \sin(ax)\,dx = -\frac{1}{a}\cos(ax)$	

Sample Exam Questions

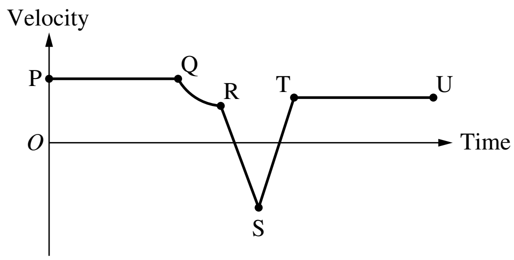
The sample exam questions that follow illustrate the relationship between the course framework and the AP Physics C: Mechanics Exam and serve as examples of the types of questions that appear on the exam. These sample questions do not represent the full range and distribution of items on an official AP Physics C: Mechanics Exam. After the sample questions is a table which shows which skill, learning objective, and essential knowledge statement each question relates to. This table also provides the answers to the multiple - choice questions.

Section I: Multiple-Choice Questions



1. A block is released from rest and slides down a track with negligible friction, descending a vertical distance of 5.0 m from its initial position to Point P, as shown in the figure. The block then slides on a horizontal surface where the coefficient of kinetic friction μ_k between the block and the horizontal surface is 0.20. How far does the block slide on the horizontal surface before coming to rest?

   (A) 1.0 m
   (B) 5.0 m
   (C) 10 m
   (D) 25 m



2. The velocity as a function of time for an object moving along a straight line is shown in the graph. For which of the following sections of the graph is the acceleration constant and nonzero?

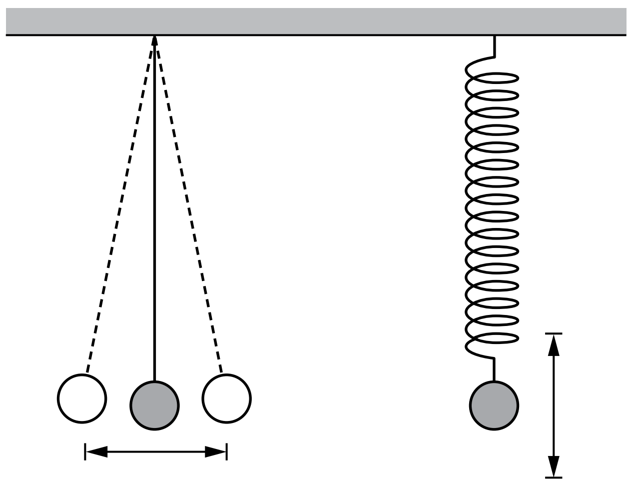
   (A) QR only
   (B) PQ and TU only
   (C) RS and ST only
   (D) PQ, RS, ST and TU only

3. The net force F exerted on an object that moves along a straight line is given as a function of time t by $F(t) = At^2 + B$, where $A = 1\ \text{N/s}^2$ and $B = 1\ \text{N}$. What is the change in momentum of the object from $t = 0$ to $t = 3\ \text{s}$?

(A) 6 kg·m/s
(B) 12 kg·m/s
(C) 17 kg·m/s
(D) 30 kg·m/s

4. A spherical star spinning at an initial angular velocity ω suddenly collapses to half of its original radius without any loss of mass. Assume the star has uniform density before and uniform density after the collapse. What is the angular velocity of the star after the collapse?

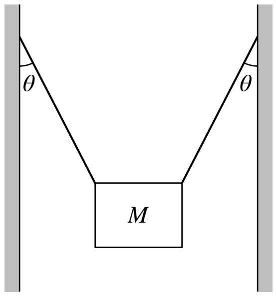
(A) $\omega / 4$
(B) $\omega / 2$
(C) 2ω
(D) 4ω



5. A pendulum is constructed by attaching a 1 kg sphere to the end of a light string that is fixed to a ceiling. A second 1 kg sphere is attached the end of a spring, which is also fixed to the ceiling. The pendulum and spring have the same length when in equilibrium, as shown in the figure. Both spheres are displaced from equilibrium in the directions shown and oscillate with the same period. If the 1 kg spheres are replaced with 2 kg spheres and the amplitudes of oscillations are unchanged, which of the following is true about each of their resulting periods of oscillation?

	Period of pendulum	Period of spring
A	Remains the same	Remains the same
B	Remains the same	Increases
C	Increases	Remains the same
D	Increases	Increases

(A) A
(B) B
(C) C
(D) D



6. A heavy sign of mass M is held at rest by two supporting wires between two buildings, with each wire making an angle θ with the vertical, as shown in the figure. What is the tension in each wire?

(A) $\dfrac{Mg}{2\sin\theta}$

(B) $\dfrac{Mg}{2\cos\theta}$

(C) $\dfrac{Mg}{\sin\theta}$

(D) $\dfrac{Mg}{\cos\theta}$

	Mass (kg)	Area (cm^2)	Coefficient of Static Friction (μ_s)
Block 1	0.75	75	0.20
Block 2	3.0	100	0.40

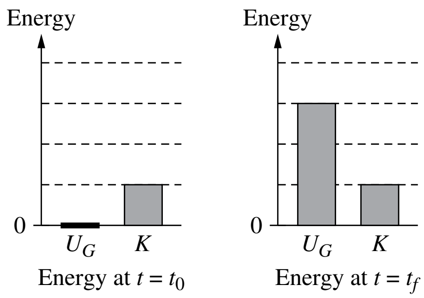
7. In an experiment, two blocks made of different materials are placed on the same wooden board, which is initially horizontal. The table shown provides three values for each block: the mass, the area of the side of the block in contact with the board, and the coefficient of static friction between the block and the board. One end of the board is then slowly raised. Which of the following correctly identifies the block that will start sliding down the board first, and includes appropriate reasoning?

(A) Block 1, because it has less mass per unit area.

(B) Block 1, because the coefficient of static friction is smaller.

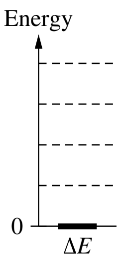
(C) Block 2, because it has greater mass.

(D) Block 2, because the coefficient of static friction is larger.

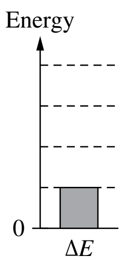


8. A student lifts a ball straight upward at a constant speed. The energy bar diagrams represent the gravitational potential energy U_G of the ball-Earth system and the kinetic energy K of the ball at a time $t = t_0$ and at a later time $t = t_f$. Which of the following energy bar diagrams correctly represents the <u>change</u> in mechanical energy ΔE of the ball-Earth system for the time interval $\Delta t = t_f - t_0$?

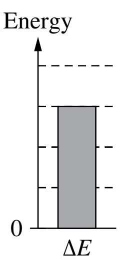
(A)



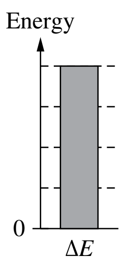
(B)

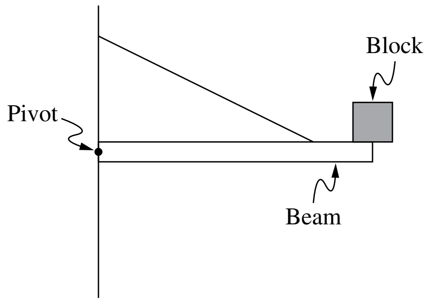


(C)



(D)





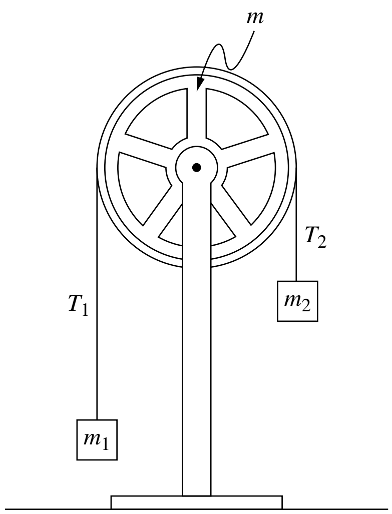
9. A uniform beam of length 0.60 m and mass 4.0 kg is fixed to a wall by a pivot. The beam is held horizontally by a string fixed to the beam and wall. A block of mass 1.0 kg is attached at the end of the beam, as shown. Which of the following is nearest to the magnitude of the torque about the pivot exerted by the weight of the block?

(A) 0

(B) 6 N·m

(C) 10 N·m

(D) 18 N·m



10. Two blocks of mass m_1 and m_2 are connected by a thin string that passes over a circular wheel of mass m, as shown. The wheel is attached to a vertical stand and rotates freely about its center. The string does not slip on the wheel and exerts forces T_1 and T_2 on the blocks. When the wheel is released from rest in the position shown, it undergoes an angular acceleration and rotates clockwise. Which of the following statements about the magnitudes of T_1 and T_2 is correct?

(A) $T_1 = T_2$ because the tension in the string must always be constant within the string.

(B) $T_1 = T_2$ because both blocks have the same acceleration.

(C) $T_1 < T_2$ because m_1 is farther from the wheel than m_2

(D) $T_1 < T_2$ because an unbalanced clockwise torque is needed to accelerate the wheel clockwise.

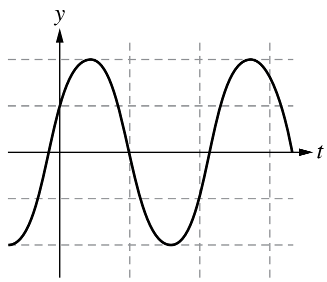
11. Two small, identical blocks, Block 1 and Block 2, are dropped from rest at different heights above a horizontal floor. Block 1 has speed v_1 immediately before hitting the floor, and Block 2 has speed $2v_1$ immediately before hitting the floor. The work done by gravity on Block 1 as it falls is W_1. What is the work done by gravity on Block 2 as it falls?

(A) $\frac{1}{4}W_1$

(B) $\frac{1}{2}W_1$

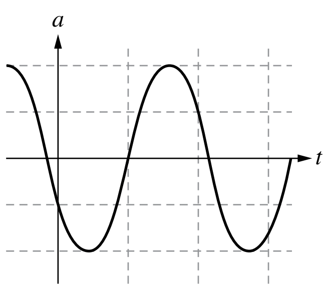
(C) $2W_1$

(D) $4W_1$

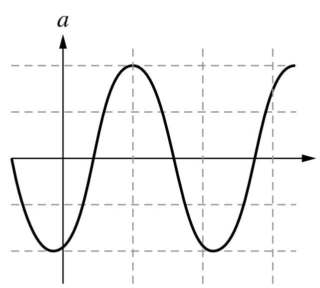


12. A block is hung vertically from an ideal spring. The block is pulled down, released, and allowed to oscillate. The vertical position y of the block as a function of time t is shown in the graph. Which of the following graphs best represents the corresponding acceleration a of the block as a function of time?

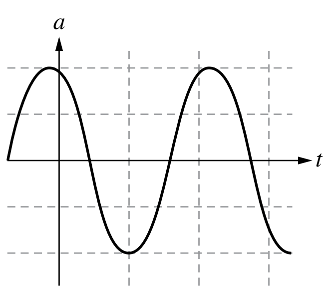
(A)



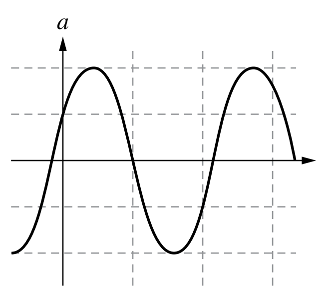
(B)



(C)



(D)



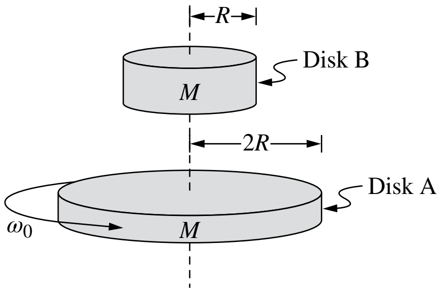
13. A small rock is launched from the surface of a planet with no atmosphere. The initial speed of the rock is $2v_{esc}$, where v_{esc} is the escape speed of the planet. What is the speed of the rock when it is very far from the surface of the planet?

(A) Zero

(B) $\frac{1}{2}v_{esc}$

(C) $\sqrt{2}v_{esc}$

(D) $\sqrt{3}v_{esc}$



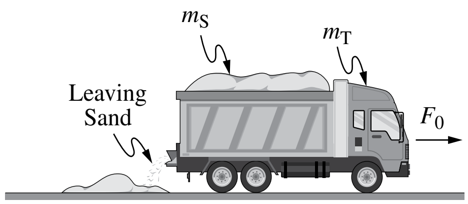
14. Two uniform disks, A and B, each of mass M, have radii $2R$ and R, respectively. The rotational inertia of a uniform disk of mass m and radius r is $\frac{1}{2}mr^2$. Disk A rotates in a horizontal plane about its center with angular velocity ω_0 and Disk B is initially held at rest. Disk B is then released and falls onto the center of Disk A such that both disks spin together about their centers without slipping. What is the angular velocity ω_f of the two-disk system?

(A) $\frac{1}{2}\omega_0$

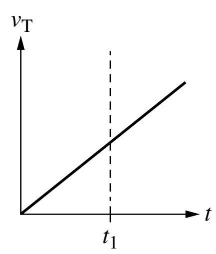
(B) $\frac{2}{3}\omega_0$

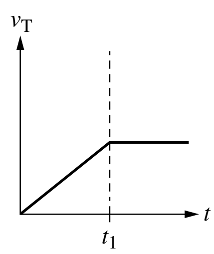
(C) $\frac{4}{5}\omega_0$

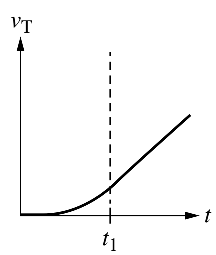
(D) $\frac{2}{\sqrt{5}}\omega_0$

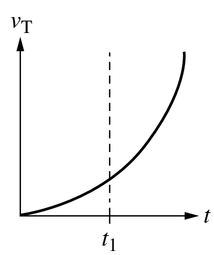


15. A truck of mass m_T is initially at rest with a load of sand of initial mass m_s. At time $t = 0$, the truck is pushed forward by a constant force F_0 and the sand begins to leave the truck, as shown. The mass of sand that has left the truck as a function of time t is given by Ct, where C is a positive constant with appropriate units. At time t_1 there is no sand left in the truck. Which of the following graphs most nearly shows the velocity v_T of the truck as a function of time?

(A)


(B)


(C)


(D)


Section II: Free–Response Questions

FREE-RESPONSE QUESTION: MATHEMATICAL ROUTINES

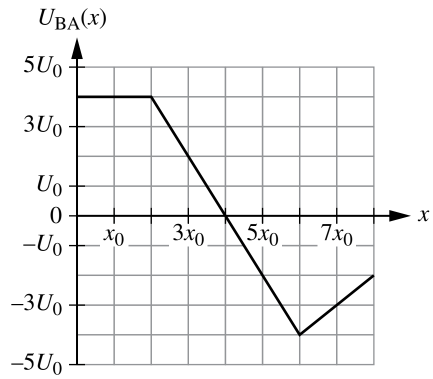


Figure 1

1. A system is comprised of two objects, A and B, which interact with each other through a conservative force $F_{BA}(x)$, where x is the position of Object A with respect to Object B. The potential energy $U_{BA}(x)$ of the two-object system as a function of x is shown in Figure 1. No external forces are exerted on the two-object system.

Object A has mass m_A and is released from rest at position $x = 3x_0$. Object A starts moving, and later passes through position $x = 7x_0$.

(a)

i. **Derive** an expression for the speed of Object A at the instant Object A is passing through position $x = 7x_0$. Express your answer in terms of m_A, U_0, x_0, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference material.

ii. On the grid shown in Figure 2, **draw** a graph of the force exerted on Object A by Object B.

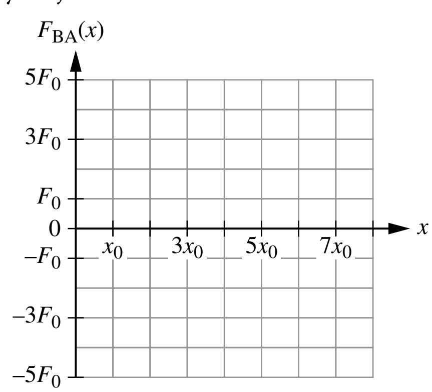


Figure 2

In another scenario, Object A is replaced by Object C. Objects C and B interact with each other through a conservative force $F_{BC}(x) = -\beta x^2$, where $\beta = \frac{3}{8}\frac{\text{kg}}{\text{s}^2\text{m}}$. The potential energy $U_{BC}(x)$ of the Object C-Object B system is defined to be zero when Object C is at position $x = -2$ m. No external forces are exerted on the two object system.

(b) **Derive** an equation for $U_{BC}(x)$. Express your answer only in terms of x. Begin your derivation by writing a fundamental physics principle or an equation from the reference material.

FREE-RESPONSE QUESTION: TRANSLATION BETWEEN REPRESENTATIONS

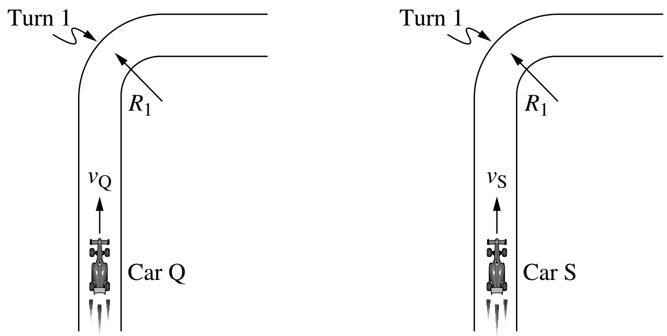
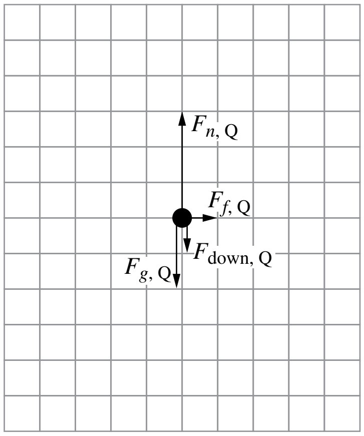


Figure 1

2. Two identical race cars Q and S, each of mass m, are driven on a flat, horizontal road. Turn 1 in the road can be modeled as a circular segment of radius R_1, as shown in Figure 1. The coefficient of static friction between each car's tires and the road is μ. The race cars are constructed so that when the cars are moving forward at a speed v relative to the air, the air exerts a downward force of magnitude F_{down} on the car given by $F_{down} = bv^2$, where b is a positive constant with appropriate units.

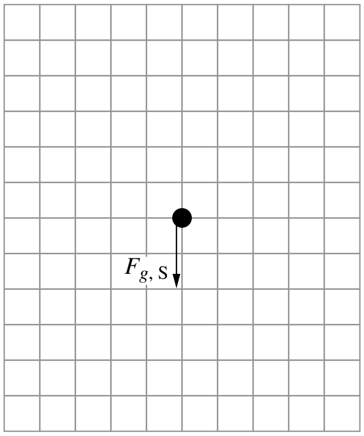
Both cars travel through Turn 1 at constant speeds without slipping. Car Q travels at speed v_Q and Car S travels at speed $v_S = 2v_Q$. The air is not moving, so the cars' speeds relative to the air is the same as their speed relative to the ground.

The dot in Figure 2 represents Car Q, as seen from the back, as the car turns to the right through Turn 1. The four arrows represent the following forces exerted on the car: $F_{down,Q}$, a gravitational force $F_{g,Q}$, a normal force $F_{n,Q}$, and a frictional force $F_{f,Q}$.



Car Q, Back View

Figure 2



Car S, Back View

Figure 3

The dot in Figure 3 represents Car S, as seen from the back, as the car turns to the right through Turn 1. The arrow labeled $F_{g,S}$ represents the gravitational force exerted on Car S.

(a) On the dot in Figure 3, **draw** and **label** the other forces exerted on Car S. Each force must be represented by a distinct arrow starting on, and pointing away from, the dot. The lengths of the arrows should reflect the relative magnitudes of the forces exerted on Car Q.

If the radius of the curve is large enough, then either car would be capable of traveling through the curve at any speed without slipping. The critical radius R_C is the smallest value of the radius of curvature where the cars can travel at any speed without slipping.

(b) **Derive** an expression for R_C in terms of m, μ, b, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference material.

The maximum speed v_{max} that a car can travel along a circular track without slipping as a function of the track's radius R if air exerts <u>no</u> downward force on the car is shown with the dashed line in Figure 4.

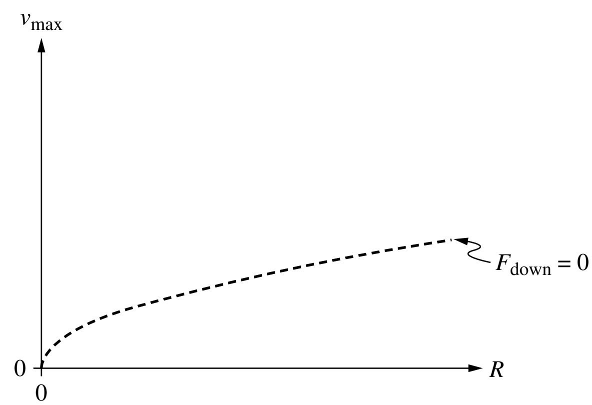


Figure 4

(c) On Figure 4, **sketch** the maximum speed that the car can have on a circular track of radius R when air exerts a downward force of magnitude $F_{down} = bv^2$ on the car.

Figure 5 shows Car Q in a semicircular turn of radius R_1. Car Q enters the turn at Point A and then travels through both Point B, which is midway through the turn, and Point C, where the car exits the turn. Air moves in the direction shown with respect to the ground at speed v_{wind}. While turning, Car Q maintains a constant radius turn and always travels at the maximum speed possible without slipping. The time intervals in which Car Q travels between points A and B and between points B and C are Δt_{AB} and Δt_{BC}, respectively.

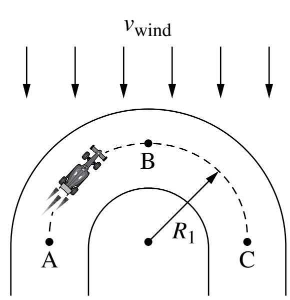


Figure 5

(d) Is Δt_{AB} is greater than, less than, or equal to Δt_{BC}? **Justify** your answer using physical principles.

FREE-RESPONSE QUESTION: EXPERIMENTAL DESIGN AND ANALYSIS

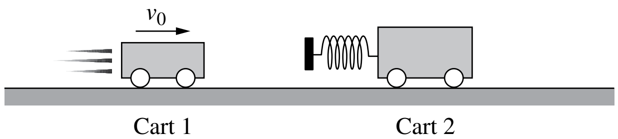


Figure 1

3. A group of students have two carts, Cart 1 and Cart 2, each of known mass M_1 and M_2, respectively. The carts are placed on a straight horizontal track. Cart 1 is given an initial speed v_0 toward the right. Cart 2 is initially at rest and has a spring attached to it, as shown in Figure 1. Cart 1 collides with Cart 2.

The group of students want to determine the relationship between v_0 and the fraction of kinetic energy remaining after the collision, which is described by the ratio $\frac{K_{\text{total},f}}{K_{\text{total},i}}$, where $K_{\text{total},f}$ is the total kinetic energy of Carts 1 and 2 after the collision and $K_{\text{total},i}$ is the total kinetic energy of Carts 1 and 2 before the collision.

(a) **Describe** an experimental procedure to collect data that would allow the students to determine if the initial speed of Cart 1 changes the ratio $\frac{K_{\text{total},f}}{K_{\text{total},i}}$. Provide enough detail so that the experiment could be replicated, including any steps necessary to reduce experimental uncertainty.

(b) **Describe** how the collected data should be analyzed to determine whether the value of v_0 changes the ratio $\frac{K_{\text{total},f}}{K_{\text{total},i}}$

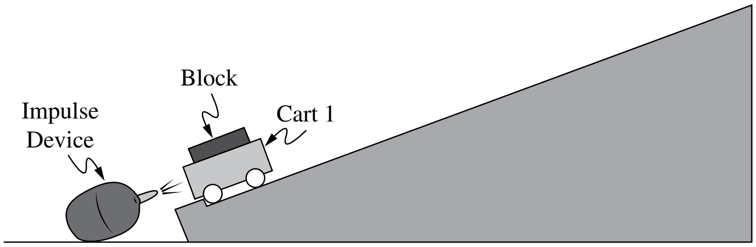


Figure 2

In a later experiment, Cart 1 is placed at the bottom of a ramp. An impulse device at the bottom of the ramp delivers an impulse to Cart 1 that is directed up the ramp by quickly releasing a burst of high-pressure air, as shown in Figure 2. The magnitude of the impulse delivered to the cart is the same in all trials. Blocks of different known masses can be attached to Cart 1.

The students are asked to determine the value of the impulse delivered to the block-cart system by the impulse device. The students measure the combined mass $M_{1,B}$ of the cart and block. The impulse device delivers the impulse to the block-cart system, and the students measure the maximum vertical height h attained by the block-cart system above the initial position of the system.

The experiment is repeated using blocks of different masses. The students' measurements are shown in the following table.

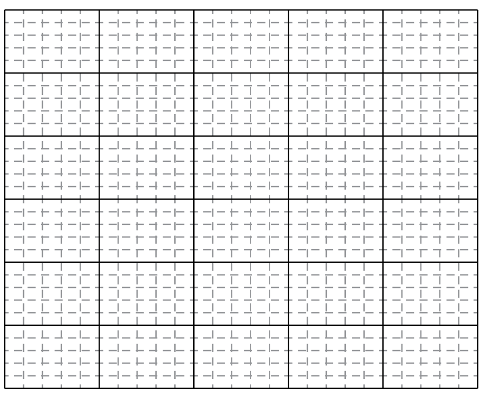
Trial Number	Combined mass of cart and block, $M_{1,B}$ (kg)	Maximum Height h (m)		
1	0.100	0.395		
2	0.150	0.160		
3	0.250	0.070		
4	0.350	0.035		
5	0.500	0.015		

(c)

i. **Indicate** two quantities that when graphed produces a straight line that could be used to determine a numerical value for the impulse delivered by the impulse device. You may use the blank columns in the table for any quantities you graph other than the given data. Use the blank columns in the table to list any calculated quantities you will graph other than the data provided.

Vertical Axis:___________ Horizontal Axis:___________

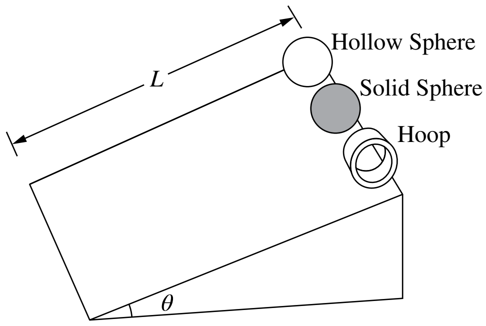
ii. **Plot** the data points for the quantities indicated in part (c)(i) on the following graph. Clearly scale and label all axes, including units.



iii. On the above graph, **draw** a straight line that best represents the data.

(d) Using the straight line from part (c)(iii), **calculate** the value of the impulse delivered to the cart by the impulse device.

FREE-RESPONSE QUESTION: QUALITATIVE/QUANTITATIVE TRANSLATION



4. A hollow sphere, a uniform solid sphere, and a hoop are placed at the top of a ramp of length L that makes an angle θ with the horizontal, as shown in the figure. The hollow sphere, solid sphere, and hoop have the same mass M and the same radius R. All three shapes are released from rest at the same time and roll down the ramp without slipping.

   (a) **Rank** the order in which the shapes reach the bottom of the ramp. A ranking of "1" indicates the shape reaches the bottom of the ramp first. If any two shapes reach the bottom of the ramp at the same time, give them the same ranking.

   _____Hollow Sphere _____Uniform Solid Sphere _____Hoop

   Justify your rankings using relevant physics principles, but without directly manipulating or deriving equations.

   The time for one of the shapes to roll without slipping to the bottom of the ramp is t, and the rotational inertia of that shape is I.

   (b) **Derive** the relationship between t and I to show that $t = \sqrt{\dfrac{2L\left(I + MR^2\right)}{MgR^2 \sin\theta}}$.

   Begin your derivation by writing a fundamental physics principle or an equation from the reference material.

   A new sphere is constructed using a thin spherical shell of radius R. The shell is completely filled with liquid. The mass of the shell is much less than that of the liquid, and the total mass of the liquid-filled sphere is M. The liquid-filled sphere is placed at the top of the ramp, released from rest, and rolls without slipping to the bottom of the ramp. As the sphere rolls down the ramp, the liquid inside the sphere does <u>not</u> rotate with the shell.

   (c) Does the liquid-filled sphere reach the bottom of the ramp in more time, less time, or the same time as the solid sphere? Use the equation derived in part (b) to **justify** your answer.

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Essential Knowledge
1	D	2.B	3.4.B	3.4.B.2
2	C	2.C	1.3.A	1.3.A.4
3	B	2.B	4.2.B	4.2.B.2
4	D	2.D	6.4.A	6.4.A.2
5	B	2.C	7.2.A	7.2.A.1
6	B	2.A	2.4.A	2.4.A.1
7	B	3.C	2.7.B	2.7.B.2
8	C	3.B	3.4.B	3.4.B.2
9	B	2.B	5.3.B	5.3.B.2
10	D	3.C	5.6.A	5.6.A.1
11	D	2.D	3.2.A	3.2.A.4
12	A	3.B	7.3.A	7.3.A.3
13	D	2.A	6.6.A	6.6.A.4
14	C	2.A	6.3.A	6.3.A.2
15	C	2.C	4.2.B	4.2.B.2

Free-Response Question	Skill	Learning Objective
1	2.A, 3.B, 1.C	3.1.A, 3.4.B, 3.3.A
2	1.A, 3.B, 2.A, 1.C, 2.D, 3.C	1.4.B, 2.2.B, 2.4.A, 2.4.B, 2.5.A, 2.7.B, 2.10.A
3	3.A, 2.D, 1.B, 2.B, 3.C	2.1.B, 3.1.A, 3.4.B, 3.4.C, 4.2.B, 4.4.A
4	3.B, 3.C, 2.A, 2.D	1.3.A, 3.4.B, 5.4.A, 5.6.A, 5.4.B, 6.1.A

AP PHYSICS C: MECHANICS

Scoring Guidelines

Scoring Guidelines for Question 1: Mathematical Routines — 10 points

Learning Objectives: 3.1.A 3.4.B 3.3.A

(a)	**i.** For a multistep derivation that indicates one of the following: • the total energy of the two object system when Object A is at position $x = 3x_0$ is equal to the total energy of the system when Object A is at position $x = 7x_0$: $E_{\text{total},0} = E_{\text{total},f}$ • that the change in kinetic energy of Object A is equal to the negative of the system's change in potential energy: $\Delta K = -\Delta U$	**1 point**
	For indicating the total energy of the two-object system is the sum of the system's potential energy and the kinetic energy of Object A: $E_{\text{total}} = U + K$	**1 point**
	For indicating that either the velocity of Object A or the kinetic energy of Object A is zero when Object A is at position $x = 3x_0$, $K_0 = 0$	**1 point**
	For correct substitutions of the potential energy of the system when Object A is at position $x = 3x_0$ and at position $x = 7x_0$: $U(7x_0) - U(3x_0) = -3U_0 - 2U_0$ OR For indicating that the change in potential energy as Object A moves from $x = 3x_0$ to $x = 7x_0$ is $-5U_0$ $\Delta U = -5U_0$	**1 point**
	For a correct answer $v = \sqrt{\dfrac{10U_0}{m_A}}$	**1 point**
	Example Response: $E_{\text{total},0} = E_{\text{total},f}$ $U_0 + K_0 = U_f + K_f$ $U_f - U_0 = -\left(K_f - K_0\right)$ $U(7x_0) - U(3x_0) = -\left(K(7x_0) - 0\right)$ $-3U_0 - 2U_0 = -\dfrac{1}{2} m_A v^2$ $v = \sqrt{\dfrac{10U_0}{m_A}}$	
(a)	**ii.** For sketching a force is equal to zero for the entire region $0 < x < 2x_0$	**1 point**
	For drawing a constant force across the interval $2x_0 < x < 6x_0$ that has twice the magnitude of a constant force across the interval $6x_0 < x < 8x_0$	**1 point**

Example Response:

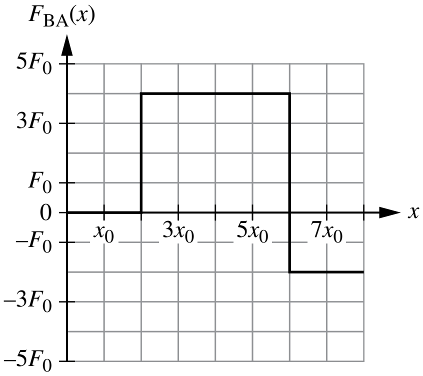


Figure 2

Total for Part (a)	**7 points**

(b) For a substituting the given expression for force into an appropriate equation to find a change in potential energy of the two-object system — **1 point**

$$\Delta U = -\int_a^b \left(\beta x^2\right) dx$$

For **one** of the following: — **1 point**

- Correctly relating the change in potential energy of the two-object system to the difference in potential energy of the system when Object A is at position x and another arbitrary position: $\Delta U = U(x) - U(x')$
- Using correct limits in the integral where a definite integral can be evaluated by setting $U(-2\text{ m}) = 0\text{ J}$
- Indicating a correct integration constant

For a correct equation for potential energy as a function of position — **1 point**

$$U(x) = \frac{x^3}{8} + 1$$

Example Response:

$$U(x) - U(-2\text{ m}) = -\int_{-2\text{ m}}^{x} F(x)\,dx$$

$$U(x) - 0 = -\int_{-2\text{ m}}^{x} \left(-\beta x^2\right) dx = \int_{-2\text{ m}}^{x} \left(\beta x^2\right) dx$$

$$U(x) = \beta\left(\frac{x^3}{3}\right)\Bigg|_{-2\text{ m}}^{x} = \left(\frac{x^3}{8}\right)\Bigg|_{-2}^{x}$$

$$U(x) = \frac{x^3}{8} - \frac{(-2)^3}{8} = \frac{x^3}{8} + 1$$

Total for Part (b)	**3 points**
Total for question 1	**10 points**

Scoring Guidelines for Question 2: Translation Between Representations — 12 points

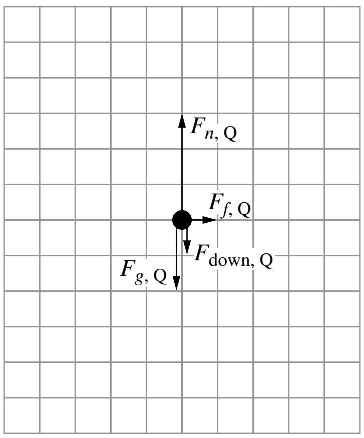
Learning Objectives: 1.4.B 2.2.B 2.4.A 2.5.A 2.7.B 2.10.A

(a) For correctly drawing and labeling the three forces $F_{\text{down,S}}$, $F_{\text{n,S}}$, and $F_{\text{f,S}}$ such that $\sum F_{\text{vertical}} = 0$ (regardless of the relative sizes of each arrow), with no extraneous forces — **1 point**

For drawing a downward force $F_{\text{down,S}} = 4F_{\text{down,Q}}$, i.e., four grid units in length — **1 point**

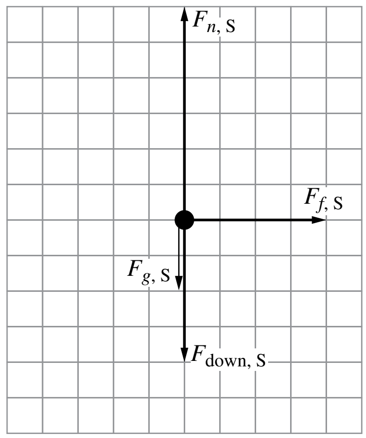
For drawing $F_{\text{f,S}}$ in the same horizontal direction as $F_{\text{f,Q}}$, with $F_{\text{f,S}} = 4F_{\text{f,Q}}$, i.e., four grid units in length — **1 point**

Example Response:



Car Q, Back View

Figure 2



Car S, Back View

Figure 3

Total for Part (a) — **3 points**

(b) For a multistep derivation that uses Newton's second law, consistent with the diagram drawn in part (a) for the vertical direction — **1 point**

$$\sum F_{\text{vert}} = 0$$

$$F_n - mg - F_{\text{down}} = 0$$

$$F_n = mg + bv^2$$

For a correct expression for the frictional force exerted on the car — **1 point**

$$F_f \le \mu F_n = \mu\left(mg + bv^2\right)$$

Scoring Note: An expression for normal force that is consistent with the previously drawn free-body diagram drawn earns this point.

For correctly equating the frictional force to the force that causes a centripetal acceleration — **1 point**

$$F_f = m\frac{v^2}{R_C}$$

For a correct expression for R_C in terms of the given variables $R_C = \dfrac{m}{\mu b}$ — **1 point**

Example Response 1:

Using Newton's second law for the vertical forces,

$$\sum F_{\text{vert}} = 0$$

$$F_n - m_C g - F_{\text{down}} = 0$$

$$F_n = m_C g + bv^2$$

The friction force is

$$F_f \le \mu F_n$$

These quantities are equal when the car is going as fast as possible without slipping, so $F_f = \mu F_n = \mu\left(m_C g + bv^2\right)$

Using Newton's second law for the horizontal friction force,

$$F_f = m_C \frac{v^2}{R_C}$$

$$\frac{m_C v^2}{R_C} = \mu\left(bv^2 + m_C g\right)$$

$$R_C = \frac{m_C v^2}{\mu\left(bv^2 + m_C g\right)}$$

As $v \to \infty$, neglect $m_C g$ so $R_C \to \frac{m_C v^2}{\mu b v^2}$, and $R_C = \frac{m_C}{\mu b}$

Example Response 2:

$$\sum F_{\text{vert}} = 0$$

$$F_n - m_C g - F_{\text{down}} = 0$$

$$F_n = m_C g + bv^2$$

$$F_f \le \mu F_n = \mu\left(m_C g + bv^2\right)$$

$$F_f = m_C \frac{v^2}{R_C}$$

$$\frac{m_C v^2}{R_C} \le \mu\left(bv^2 + m_C g\right)$$

$$v^2\left(\frac{m_C}{R_C} - \mu b\right) \le \mu m_C g$$

$$v^2 \le \frac{\mu m_C g}{\frac{m_C}{R_C} - \mu b}$$

v can have any value when the denominator goes to zero, so

$$\frac{m_C}{R_C} - \mu b = 0$$

$$\frac{m_C}{R_C} = \mu b$$

$$R_C = \frac{m_C}{\mu b}$$

Total for Part (b) **4 points**

(c)	For a sketch that is greater than the given curve for all $R > 0$	**1 point**
	For a sketch that starts at the origin and is always increasing	**1 point**
	For a sketch that has a single transition from concave down to concave up **Scoring Note:** A vertical asymptote is not required to earn this point.	**1 point**

Example Response:

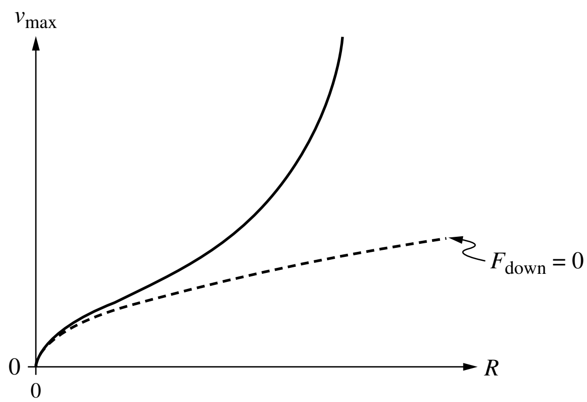


Figure 4

Total for Part (c) **3 points**

(d)	For indicating that the maximum speed before slipping is either greatest when the car and wind travel in opposite directions, or equivalently, least when the car and wind travel in the same direction.	**1 point**
	For a justification that correctly relates average speed to the time taken to travel between two points	**1 point**

Example Response:

The relative speed between the car and the air is greater when the car is moving into the wind between Point A *and Point* B*, so the downward force and maximum speed are greater as well. Since the car can go faster between points* A *and* B *than between points* B *and* C*, the car spends less time between points* A *and* B*. Therefore* $\Delta t_{AB} < \Delta t_{BC}$.

Total for part (d) **2 points**

Total for Question 2 **12 points**

Scoring Guidelines for Question 3: Experimental Design and Analysis — 10 points

Learning Objectives: 2.1.B 3.1.A 3.4.B 3.4.C 4.2.B 4.4.A

(a) For repeating the experiment for a range of initial velocities of Cart 1 — **1 point**

Scoring Note: While best practice encourages students to also perform multiple trials using the same v_0, these measurements are not required to earn this point.

For measuring the speed of Cart 1 before the collision and both carts after the collision — **1 point**

Example Response:

1. *Position motion sensors at each end of the track.*
2. *Use the motion sensors to measure the initial speed of Cart 1 and the final speeds of both carts.*

Repeat steps 1 and 2 for multiple values of v_0

Total for Part (a) — **2 points**

(b) For indicating that the kinetic energy of the system before and after the collision will be calculated — **1 point**

For using the ratio between the initial and final kinetic energies of the system to determine if the ratio $\dfrac{K_{\text{total},f}}{K_{\text{total},i}}$ changes at different values of v_0 — **1 point**

Example Response

Use the masses and speeds of the carts before and after the collision to calculate the total kinetic energy of the two carts before ($K_{\text{total},i}$) *and after* ($K_{\text{total},f}$) *the collision. Then graph final kinetic energy as a function of initial kinetic energy. The ratio* $\dfrac{K_{\text{total},f}}{K_{\text{total},i}}$ *is equal to the slope of a line of best fit for this graph. If the data is linear, then the ratio is constant for all values of* v_0.

Total for Part (b) — **2 points**

(c) **i.** For indicating two quantities that have a linear relation and can be used to determine the impulse delivered by the impulse device — **1 point**

Scoring Note: Examples include

- $\dfrac{1}{M_{1,\text{B}}}$ and $\sqrt{h}$ (on either axis)
- $M_{1,\text{B}}^2$ and $\dfrac{1}{h}$ (on either axis)

ii. For graphing axes that have: — **1 point**

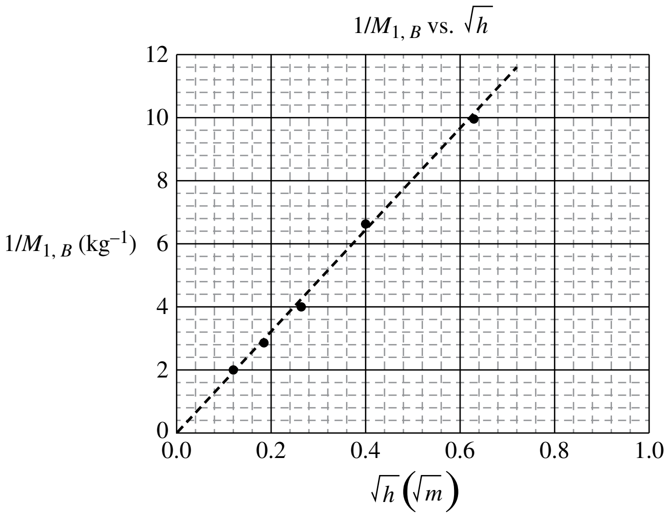
- Linear scales for which the data uses at least half the axis range
- Units consistent with the quantities listed in part (c)(i)

For correctly plotting data values from the table that are consistent with part (c)(i) — **1 point**

iii. For drawing a line that approximates the trend of the data — **1 point**

Example Response:

Trial Number	Total Mass $M_{1,B}$ of block-Cart 1 (kg)	Maximum Height h (m)	$1/M_{1,B}$ (kg^{-1})	$\sqrt{h}$ $(\sqrt{m})$	
1	0.100	0.395	10	0.628	
2	0.150	0.160	6.67	0.4	
3	0.250	0.070	4	0.265	
4	0.350	0.035	2.85	0.187	
5	0.500	0.015	2	0.122	



Total for Part (c) **4 points**

(d) For correctly relating the slope of the graphed line to the impulse **1 point**

For a value of the impulse delivered by the impulse device that is between 0.24 N · s and 0.30 N · s **1 point**

Example Response

$J = \Delta p$

$J = M_{1,B} v_0$

$v_0 = \sqrt{2gh}$

$J = M_{1,B}\sqrt{2gh}$

$$\frac{1}{M_{1,B}} = \frac{\sqrt{2g}}{J}\sqrt{h}$$

$$\text{slope} = \frac{\sqrt{2g}}{J}$$

$$J = \frac{\sqrt{2g}}{\text{slope}}$$

$$J = \frac{\sqrt{2g}}{16.16} = 0.27\ \text{N}\cdot\text{s}$$

Total for part (d) **2 points**

Total for Question 3 **10 points**

Scoring Guidelines for Question 4: Qualitative/Quantitative Translation — 8 points

Learning Objectives: 1.3.A 3.4.B 5.4.A 5.6.A 5.4.B 6.1.A

(a)	For a correct ranking: Hollow sphere – 2; Solid Sphere – 1; Hoop – 3	**1 point**
	For a justification that includes one of the following: • Relating the amount of potential energy transformed into rotational kinetic energy to the rotational inertia of the shape • Relating the time the shapes take to reach the bottom of the ramp to rotational inertia	**1 point**
	For a justification that includes one of the following: • Relating the amount of translational kinetic energy of the shape to the speed of the shape at the bottom of the ramp • Relating torque exerted on and angular acceleration of the shape to the time the shapes take to reach the bottom of the ramp	**1 point**
	Example Response: *Hollow sphere – 2; Solid Sphere – 1; Hoop – 3* *The more rotational inertia the object has, the more gravitational potential energy is converted into rotational kinetic energy, leaving less translational kinetic energy. The least kinetic energy will move the slowest and reach the bottom of the ramp last. Because an object's rotational inertia is related to the average distance of the object's mass from its rotational axis, the hoop will move the slowest and the solid sphere the fastest.*	
	Total for Part (a)	**3 points**
(b)	For a multistep derivation that uses Newton's second law in rotational form to determine a translational acceleration $\tau_{\text{net}} = I\alpha$ $\alpha = \dfrac{a}{R}$	**1 point**
	For a net torque and rotational inertia of the sphere that are consistent with the chosen axis of rotation about which to analyze the motion of the sphere: $MgR\sin\theta = \left(I + MR^2\right)\alpha$ (gravity exerts a torque on the sphere about the point of contact between the sphere and ramp) *OR* $F_f R = I\alpha$ and $Mg\sin\theta - F_f = Ma$ (friction exerts a torque on the sphere about the sphere's center of mass) **Scoring Note:** A derivation that correctly applies conservation of energy can earn points 1 and 2.	**1 point**
	For using an appropriate equation that relates translational acceleration, distance, and time $x = x_0 + v_0 t + \dfrac{1}{2}at^2$	**1 point**

Example Response:

$\tau_{\text{net}} = I\alpha$

$MgR\sin\theta = \left(I + MR^2\right)\alpha$

$MgR\sin\theta = \left(I + MR^2\right)\left(\dfrac{a}{R}\right)$

$a = \dfrac{MgR^2\sin\theta}{I + MR^2}$

$x = x_0 + v_0 t + \dfrac{1}{2}at^2$

$L = \dfrac{1}{2}\left(\dfrac{MgR^2\sin\theta}{I + MR^2}\right)t^2$

$t = \sqrt{\dfrac{2L\left(I + MR^2\right)}{MgR^2\sin\theta}}$

	Total for Part (b)	**3 points**
(c)	For correctly relating the amount of rotational inertia of the shell to the time for the shell to roll down the ramp	**1 point**
	For indicating that the liquid-filled sphere will have a smaller rotational inertia than the solid sphere because less mass is rotating as the liquid-filled sphere rolls down the ramp	**1 point**

Example Response:

Because the angle, length of the ramp, mass, and radius of the shell is the same as the solid sphere, the derivation in part (b) indicates the time to roll down the ramp will only change if the rotational inertia changes. The liquid-filled sphere must have a smaller rotational inertia because less mass will rotate as the sphere rolls down the ramp. Therefore t will be smaller because I is smaller.

Total for Part (c)	**2 points**
Total for Question 4	**8 points**